

Kapil Sandhu

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Research Profile

Computer Science Master's student at the University of Bonn with a strong interest in deep learning and sequence modeling. Seeking a Student Assistant position in Prof. Flek's group (supervised by Dr. João A. Gonçalves) to work on large language models and foundation models for physics. Experienced with Transformer architectures, neural sequence models, and simulation development.

Education

University of Bonn

MSc. Computer Science

Apr 2026 – Present

Bonn, Germany

Indian Institute of Information Technology Sonipat

BTech in Computer Science and Engineering (Grade: 1.9)

Oct 2021 – Jun 2025

Sonipat, India

Selected Projects

Fine-Tuning Causal Language Models for Natural Language Inference 🐙 causal-nli Python, PyTorch, Hugging Face, DDP, LoRA, EMA, Accelerate

- Fine-tuned causal architectures (0.6B and 1.7B parameters) on 570,000 sentence pairs from the SNLI dataset using LoRA (bfloat16 mixed precision) and gradient checkpointing.
- Set up multi-GPU training using PyTorch Distributed Data Parallel (DDP) and stabilized weight updates with Exponential Moving Average (EMA).
- Ran ablation studies on training optimization techniques and evaluated model performance (reaching 90.43% accuracy on SNLI) and robustness on Multi-NLI and adversarial datasets.

Agentic Trade Analyzer 🐙 agentic-trade-analyzer

Python, PyTorch, LangChain, Gemini API, Redis, WebSockets

- Developed a Mixture-of-Experts (MoE) multi-agent quantitative trading analyzer using Gemini 2.5 Flash.
- Implemented a 3-round adversarial debate protocol between Bull and Bear LLM agents to validate trade signals.
- Built a real-time market microstructure engine over Binance WebSockets, streaming EWMA order-flow imbalances via Redis.

CPro - Multi-time frame Market Regime Detector 🐙 cpro

Python, PyTorch, Transformers, SciPy

- Built a market regime detector for Bitcoin using a Transformer model that processes multiple timeframes (1m, 5m, 15m, 1h).
- Used multi-head self-attention to learn sequence patterns, reaching a 64% directional accuracy and improving overall model performance by 11%.

Autonomous Vehicle Localization and Tracking 🐙 Park-Vision-AI

Python, PyTorch, YOLO, OpenCV, EasyOCR, NumPy, SciPy

- Developed an autonomous vehicle localization system integrating custom CNNs with YOLO for real-time bounding-box detection.
- Implemented the Simple Online and Realtime Tracking (SORT) algorithm with Kalman filters for robust multi-object temporal tracking.
- Built a multimodal OCR extraction pipeline using EasyOCR and processed spatial bounding coordinate calculations via NumPy matrix operations, achieving an 89% tracking accuracy.

Professional Experience

Bluestock Fintech – Software Development Engineer Intern

Feb 2025 – Jun 2025

- Built fintech features like Stock Scanner and IPO management tools, and integrated ML models for market analytics.
- Worked on optimizing data processing using Node.js and TensorFlow to lower database query latencies.

Technical Skills

ML/DL/NLP Frameworks: PyTorch, TensorFlow, Hugging Face (Transformers), MLflow, scikit-learn, YOLO

Programming Languages: Python, C, C++, Java, JavaScript, CUDA, SQL, HTML/CSS, Dart

Cloud & Infrastructure: AWS, Docker, SLURM, PySpark, Git, Flutter

Web & Databases: FastAPI, React, Node.js, MongoDB, MySQL

Core Expertise: Natural Language Processing (NLP), Sequence Modeling, Foundation Models (FM), Object Detection, Computer Vision, AI agents

Languages, Honours & Awards

Languages: English (Fluent - C1), German (Basic - A2, actively improving to B1), Hindi (Native)

JEE Mains & Advanced 2021: Cleared competitive national entrance exams (top 2% nationally), achieving category rank 1758 in JEE Advanced.

German Language Course A1–A2: Completed certification for basic workplace communication and integration.